

Covariance

SAT scores

Background

You are given data on the SAT reading and writing scores of several students from our lesson on cross tables and scatter plots

Task 1

Determine if this is sample or population

Task 2

Calculate the covariance of the two datasets

Task 3

Plot the data on scatter plot and using your previous knowledge comment on whether there is a noticeable relationship between the two variables.

Solution:

Writing	Reading	$(x-\bar{x})*(y-\bar{y})$
344	378	19,490.16
383	349	19,004.16
611	503	1,179.36
713	719	44,714.16
536	503	234.36
Mean	517	490
		Sum
		Sample size
		Cov. Sample

Task 1: This is obviously a sample of only 5 students.

Task 2: The covariance is 21,155.55

Task 3: You can get two scatter plots, which look in the following ways. There is a noticeable relationship between the two variables.

